

Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details

Batch Number: 1

Start date : 02-07-2026

Names:

- 1 Gopiseti Prudhvi
- 2 Arya Teja
- 3 Ankit Shaw
- 4 Tanmyee Chandajkar
- 5 Aliya Nissam
- 6 Akshata

Internship Duration: 40-day (8 Weeks)

1. Project Title

AI-Based-Industrial-Maintenance-and-Process-Optimization

2. Project Objective

To develop an AI-based predictive maintenance system that can predict potential industrial machine failures before they occur.

To analyze machine operating parameters such as temperature, rotational speed, torque, and tool wear to identify conditions associated with machine failures.

To develop and compare machine-learning models such as Logistic Regression, Decision Tree, Random Forest, and XGBoost for accurate failure prediction.

To identify the most important factors affecting machine failure using feature-importance analysis.

To optimize industrial processes by identifying high-risk operating conditions and providing data-driven maintenance insights.

To reduce unexpected machine downtime and maintenance costs by supporting timely preventive maintenance decisions.

To provide a user-friendly dashboard for monitoring machine conditions, viewing model performance, and predicting machine failure risk.

3. Project description in detail

The AI-Based Industrial Maintenance and Process Optimization project is a machine-learning-based system developed to predict industrial machine failures before they occur and identify the machine operating conditions that contribute to failures. The project uses industrial machine data containing parameters such as air temperature, process temperature, rotational speed, torque, tool wear, and machine failure information.

The project begins by collecting and preprocessing the industrial dataset. Unnecessary data is removed, missing or duplicate values are handled, and categorical data is converted into a suitable numerical format. Exploratory Data Analysis is then performed using graphs and statistical analysis to understand the relationship between machine operating parameters and failures.

Multiple machine-learning classification models, including Logistic Regression, Decision Tree, Random Forest, and XGBoost, are trained using the prepared data. The models are evaluated using Accuracy, Precision, Recall,

F1-score, Training Time, and Confusion Matrix. The models are compared to identify the most suitable model for predicting machine failures.

The selected model is then used to predict whether a machine is likely to experience a failure based on its current operating conditions. The system can also calculate the probability of failure, allowing high-risk machines to be identified for inspection or preventive maintenance.

The project also performs feature importance analysis to determine which machine parameters have the greatest influence on failure predictions. These results are used for process optimization, where high-risk operating conditions can be identified and maintenance or process adjustments can be recommended.

Finally, the complete system can be presented through a Streamlit dashboard. The dashboard displays machine data analysis, failure statistics, model performance, confusion matrix, important machine parameters, and a prediction interface where users can enter machine conditions and obtain a failure-risk prediction.

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4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Internship orientation and Python basics	Completed orientation and learned Python fundamentals
Week 2	NumPy and numerical operations	Practiced NumPy arrays, indexing, slicing, and operations
Week 3	Pandas and data manipulation	Practiced DataFrames, data cleaning, filtering, and analysis
Week 4	Matplotlib data visualization	Created different charts and plots using Matplotlib
Week 5	Seaborn data visualization	Created count plots, box plots, bar plots, and heatmaps
Week 6	Machine Learning and Logistic Regression	Implemented Logistic Regression for classification
Week 7	Decision Tree algorithm	Implemented Decision Tree classification
Week 8	Random Forest algorithm	Implemented Random Forest and learned ensemble learning

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Defined project objectives, scope, responsibilities, and success criteria.	09 July 2026
Prototype / First Draft	Developed initial working prototype covering resume parsing, matching logic, and UI flow.	022 July 2026
Mid-Term Review	Conducted review to evaluate progress, refine feature priorities, and implement improvement actions.	06 August 2026
Final Submission	Completed and submitted the fully functional SkillMatch system with reporting and documentation.	24 August 2026
Final Presentation	Delivered demonstration highlighting system architecture, functionality, results, and impact.	27 August 2026

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5b. Project execution details

The execution of the **AI-Based Industrial Maintenance and Process Optimization** system followed a structured and iterative development approach. The project commenced with **requirement analysis**, defining the project objectives, problem scope, and identifying the applicability of AI/ML techniques to industrial predictive maintenance. The **AI4I 2020 Predictive Maintenance Dataset** was used to simulate real-world industrial machine conditions, including parameters such as air temperature, process temperature, rotational speed, torque, tool wear, and machine failure.

Next, **data preprocessing and analysis** modules were implemented. The industrial dataset was loaded and analyzed using Python, Pandas, and NumPy. Data cleaning, duplicate removal, categorical data encoding, and feature selection were performed to prepare the dataset for machine-learning models. Exploratory Data Analysis was carried out using **Matplotlib and Seaborn** to identify relationships between machine operating parameters and failure conditions through statistical graphs, distributions, and correlation analysis.

A **machine-failure prediction system** was developed using multiple classification algorithms, including **Logistic Regression, Decision Tree, Random Forest, and XGBoost**. The prepared dataset was divided into training and testing sets, and the models were trained to predict whether a machine was likely to experience failure. Model performance was evaluated using **Accuracy, Precision, Recall, F1-score, Training Time, and Confusion Matrix**. The models were then compared to identify the most suitable model for predictive maintenance.

A **feature-importance analysis** was also implemented to identify the machine parameters that have the greatest influence on failure predictions. These insights were used for **process optimization**, helping identify high-risk operating conditions and supporting data-driven maintenance decisions. The prediction results can be used to prioritize machines

for inspection and preventive maintenance before unexpected failures occur.

The system underwent iterative testing using different machine conditions and test data to validate data preprocessing, prediction accuracy, model performance, and failure-risk identification. Errors were analyzed and corrected, and the machine-learning workflow was refined to improve the reliability of the results.

Finally, the project was refined, documented, and presented as an **AI-based industrial maintenance and process optimization solution**. The completed system demonstrates how machine-learning techniques can support **early machine-failure prediction, preventive maintenance, identification of important failure factors, reduction of unexpected downtime, and improvement of industrial process efficiency**, achieving the intended objectives of the project.

6. Snapshots / Screenshots

AI-Based-Industrial-Maintenance-and-Process-Optimization-main

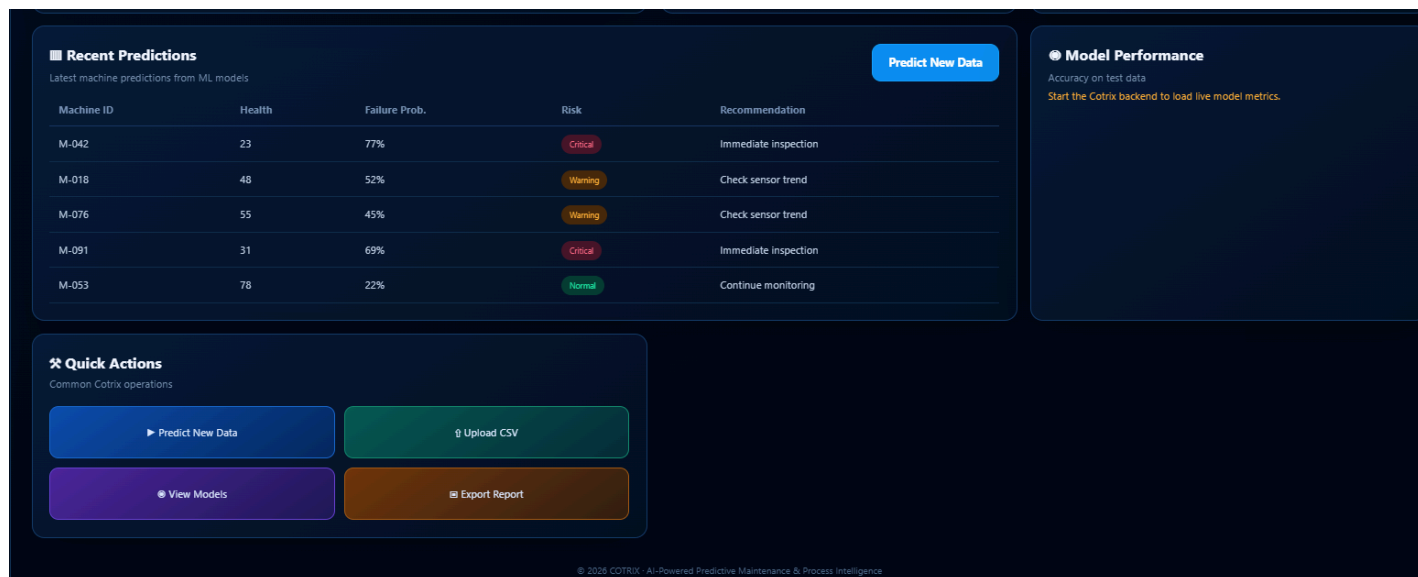
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Milestone_2	File folder			
Milestone_3	File folder			
Milestone_4	File folder			
ai4i2020	Comma Separated Values ...	135 KB	No	510 KB
LICENSE	File	1 KB	No	2 KB
README.md	MD File	4 KB	No	8 KB

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1 |<doctype html>
2 |<html lang="en">
3 |<head>
4 |<meta charset="utf-8">
5 |<meta name="viewport" content="width=device-width,initial-scale=1">
6 |<title>COTRIX – Predictive Maintenance Intelligence</title>
7 |<style>
8 |:root{--bg:#020817;--panel:#07162c;--panel2:#0a1d39;--line:#173761;--text:#f3f7ff;--muted:#8da4c3;--blue:#1595ff;--cyan:#20d8ff;--green:#21e0a0;--amber:
9 |*{box-sizing:border-box}body{margin:0;background:radial-gradient(circle at 20% 10%,#0b2850 0,#031026 28%,#020817 65%);color:var(--text);font-family:Inter,Segoe UI,A
10 |body:before{content:"";position:fixed;inset:0;background:linear-gradient(90deg,#1a1431,.98),#0b2850 0,#031026 28%,#020817 65%);color:var(--text);font-family:Inter,Segoe UI,A
11 |.app{position:relative;min-height:100vh;display:grid;grid-template-columns:220px 1fr.side(border-right:1px solid #173761;background:#0b2850 0,#031026 28%,#020817 65%);padding:22px 12
12 |@media(max-width:1100px){.kpis{grid-template-columns:repeat(3,1fr)}.grid{grid-template-columns:1fr 1fr}.wide{grid-column:span 2}}@media(max-width:760px){.app{grid-templat
13 |</style>
14 |</head>
15 |<body>
16 |<div class="app">
17 |<aside class="side">
18 |<div class="logo"><div class="logoMark"></div><div class="logoName">COTRIX</div><div class="logoSub">Industrial Intelligence</div></div></div>
19 |<nav class="nav">
20 |<button class="active" data-page="overview"><span>Overview</span></button>
21 |<button data-page="machines"><span>Machines</span></button>
22 |<button data-page="predictions"><span>Predictions</span></button>
23 |<button data-page="models"><span>ML Models</span></button>
24 |<button data-page="alerts"><span>Alerts</span><b class="badge">5</b></button>
25 |<button data-page="data"><span>Data & Reports</span></button>
26 |</nav>
27 |<div class="sideBottom"><div class="online"><i></i><b>System Online</b></div><div style="margin-top:4px">Last sync: 2 min ago</div><div class="user"><div class="avatar">
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32 |<div class="content">
33 |<section class="hero"><div class="heroText"><div class="eyebrow">PREDICTIVE MAINTENANCE</div><h1>Know what will fail <em>before</em> it does.</h1><p>Cotrix turns industri
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35 |<section id="overview" class="page active">
36 |<div class="titlebar"><div><h2>Operations Overview</h2><p>Live system status and key metrics</p></div><div class="actions"><button class="btn" onclick="toast('Dashboard r
37 |<div class="kpis">
38 |<div class="kpi"><span class="label">Machines Monitored</span><div class="value">124</div><small>↑ 8 this month</small></div>
39 |<div class="kpi"><span class="label">Fleet Health</span><div class="value">94</div><small>↑ 3.2% vs last week</small></div>
40 |<div class="kpi"><span class="label">Healthy Machines</span><div class="value">101</div><small>81.5% of fleet</small></div>
41 |<div class="kpi"><span class="label">At-Risk Machines</span><div class="value">18</div><small>14 since yesterday</small></div>
42 |<div class="kpi"><span class="label">Critical Alerts</span><div class="value">5</div><small>1 Require action</small></div>
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44 |<div class="charts">
45 |<div class="chart"><h3>Fleet Health Trend</h3><p>Average health score across monitored assets</p><div>
46 |<div class="chart"><h3>Machine Health Distribution</h3><p>Current machine state</p><div>
47 |<div class="chart"><h3>Live Alerts</h3><p>Latest machine events</p><div>
48 |<div class="alert"><div><div>Machine #042</div><div>Critical Temperature</div><div>2 min ago</div></div>
49 |<div class="alert"><div><div>Machine #018</div><div>High Vibration Detected</div><div>15 min ago</div></div>
50 |<div class="alert"><div><div>Machine #076</div><div>Tool Wear High</div><div>1 hour ago</div></div>
51 |<div class="alert"><div><div>Machine #053</div><div>Maintenance Due</div><div>3 hours ago</div></div>
52 |<div class="alert"><div><div>Machine #091</div><div>Pressure Anomaly</div><div>5 hours ago</div></div>
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pg. 4

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```

6  import numpy as np
7  import pandas as pd
8  import joblib
9  import matplotlib.pyplot as plt
10 from sklearn.svm import SVC
11 from sklearn.metrics import (
12     accuracy_score, confusion_matrix, classification_report,
13     ConfusionMatrixDisplay, roc_auc_score, average_precision_score
14 )
15
16 # =====
17 # 1. Load Preprocessed Datasets (Milestone_1)
18 # =====
19 print("Loading preprocessed data from Milestone_1...")
20 X_train = pd.read_csv('Milestone_1/X_train_scaled.csv')
21 X_test = pd.read_csv('Milestone_1/X_test_scaled.csv')
22 y_train = pd.read_csv('Milestone_1/y_train.csv').values.ravel()
23 y_test = pd.read_csv('Milestone_1/y_test.csv').values.ravel()
24
25 print(f"Train shape: {X_train.shape}, Test shape: {X_test.shape}")
26 print(f"Class distribution in y_train: {dict(zip(*np.unique(y_train, return_counts=True)))}")
27
28 # =====
29 # 2. Train SVM Model (RBF Kernel)
30 # =====
31 print("\nTraining SVM model (RBF kernel)...")
32 svm = SVC(kernel='rbf', C=1.0, gamma='scale', class_weight='balanced',
33           probability=True, random_state=42)
34 svm.fit(X_train, y_train)
35 print("SVM Model Trained Successfully!")
36

```

```
6  import numpy as np
7  import pandas as pd
8  import joblib
9  import matplotlib.pyplot as plt
10 import seaborn as sns
11 from sklearn.tree import DecisionTreeClassifier, plot_tree
12 from sklearn.metrics import (
13     classification_report, confusion_matrix,
14     roc_auc_score, ConfusionMatrixDisplay, average_precision_score
15 )
16
17 # =====
18 # 1. Load Preprocessed Datasets (Milestone_1)
19 # =====
20 print("Loading preprocessed data from Milestone_1...")
21 X_train = pd.read_csv('Milestone_1/X_train_scaled.csv')
22 X_test  = pd.read_csv('Milestone_1/X_test_scaled.csv')
23 y_train = pd.read_csv('Milestone_1/y_train.csv').values.ravel()
24 y_test  = pd.read_csv('Milestone_1/y_test.csv').values.ravel()
25
26 print(f"Train shape: {X_train.shape}, Test shape: {X_test.shape}")
27 print(f"Class distribution in y_train: {dict(zip(*np.unique(y_train, return_counts=True)))}")
28
29 # =====
30 # 2. Train Decision Tree Model
31 # =====
32 print("\nTraining Decision Tree Classifier...")
33 model = DecisionTreeClassifier(
34     max_depth=6,
35     class_weight='balanced',
36     random_state=42
37 )
```



```

7     import os
8     import numpy as np
9     import pandas as pd
10    import matplotlib.pyplot as plt
11    import seaborn as sns
12    import warnings
13    warnings.filterwarnings("ignore")
14
15    from sklearn.neural_network import MLPClassifier
16    from sklearn.metrics import (
17        precision_score, recall_score, f1_score,
18        average_precision_score, classification_report,
19        confusion_matrix, precision_recall_curve,
20        roc_curve, auc, accuracy_score
21    )
22    import joblib
23
24    OUTPUT_DIR = "Milestone_3"
25    os.makedirs(OUTPUT_DIR, exist_ok=True)
26
27    # =====
28    # 1. Load Preprocessed Datasets (Milestone_1)
29    # =====
30    print("Loading preprocessed data from Milestone_1...")
31    X_train = pd.read_csv("Milestone_1/X_train_scaled.csv")
32    X_test = pd.read_csv("Milestone_1/X_test_scaled.csv")
33    y_train = pd.read_csv("Milestone_1/y_train.csv").values.ravel()
34    y_test = pd.read_csv("Milestone_1/y_test.csv").values.ravel()
35
36    print(f"X_train shape : {X_train.shape}")
37    print(f"X_test shape : {X_test.shape}")
38    print(f"y train distribution: {dict(zip(*np.unique(y_train, return counts=True)))}")

```

```
1     import pandas as pd
2     import matplotlib.pyplot as plt
3     import seaborn as sns
4
5     from xgboost import XGBClassifier
6     from sklearn.model_selection import train_test_split
7     from sklearn.metrics import (
8         accuracy_score,
9         precision_score,
10        recall_score,
11        f1_score,
12        confusion_matrix,
13        classification_report
14    )
15
16    # -----
17    # 1. Load AI4I 2020 Predictive Maintenance Dataset
18    # -----
19
20    df = pd.read_csv("ai4i2020.csv")
21
```

```

=== Confusion Matrix (Threshold 0.50) ===
[[1596  336]
 [  12   56]]

=== Classification Report ===
              precision    recall  f1-score   support

     0       0.99      0.83      0.90      1932
     1       0.14      0.82      0.24         68

   accuracy          0.83      2000
  macro avg          0.57      2000
 weighted avg          0.96      2000

ROC-AUC Score: 0.9069

=== Results with Custom Threshold (0.3) ===
[[1310  622]
 [   5   63]]
              precision    recall  f1-score   support

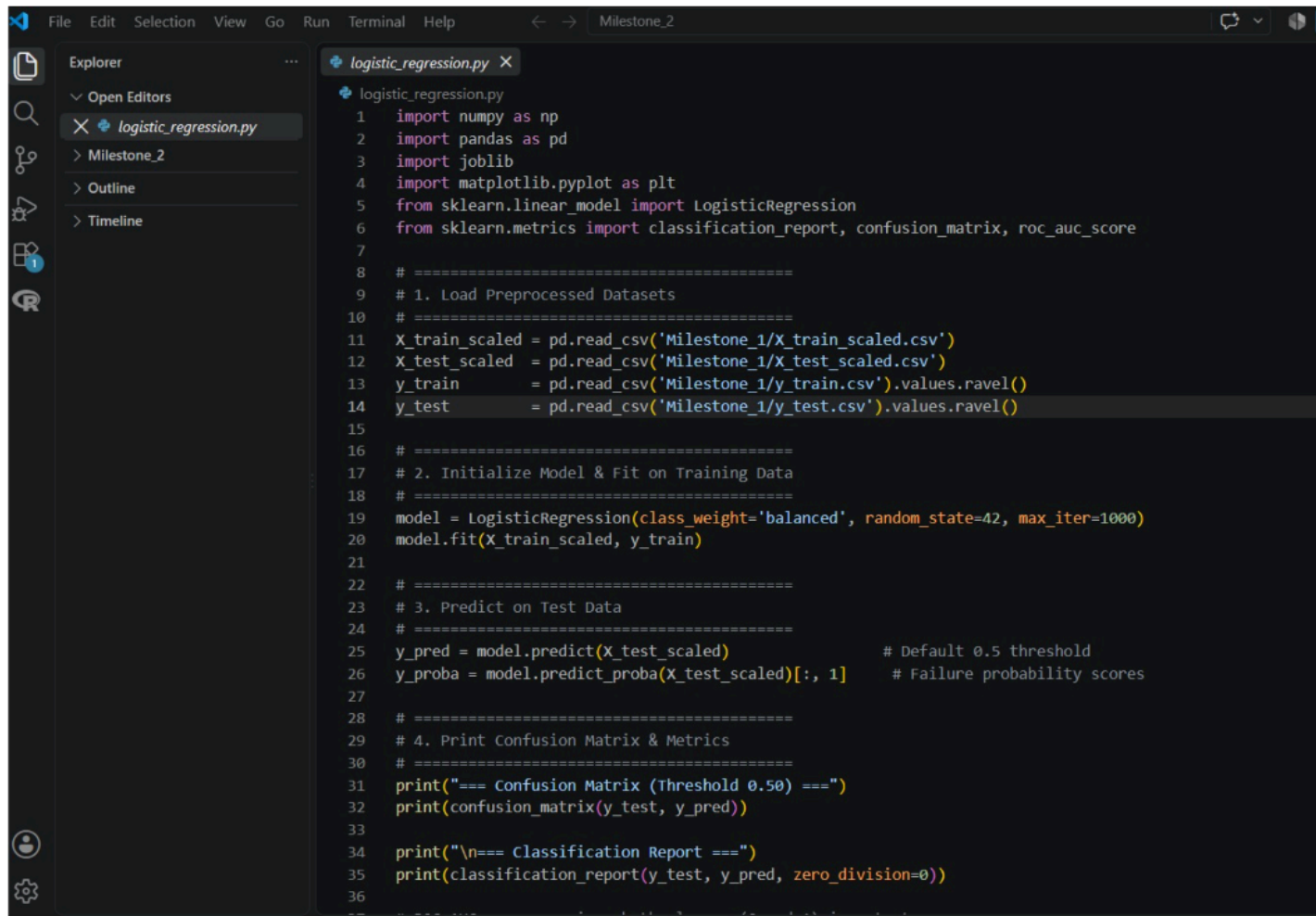
     0       1.00      0.68      0.81      1932
     1       0.09      0.93      0.17         68

   accuracy          0.69      2000
  macro avg          0.54      2000
 weighted avg          0.97      2000

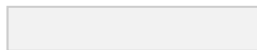
=== Top 10 Highest Risk Predictions ===
Actual Label Predicted Label Failure Risk (%)
          1           1          99.67
          1           1          98.96
          1           1          98.92
          1           1          98.68
          1           1          98.36
          1           1          98.26
          0           1          98.26
          1           1          98.03
          0           1          97.25
          0           1          97.16

```

```
Random Forest (RandomForestClassifier) Gradient Boosting (GradientBoostingClassifier) .py X ensemble_models.py X roc_curves.png
Milestone_3 > ensemble_models.py
1 import os
2 import joblib
3 import numpy as np
4 import pandas as pd
5 import matplotlib.pyplot as plt
6 import seaborn as sns
7
8 # ML Models
9 from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
10
11 # Resampling & Tuning
12 from imblearn.over_sampling import SMOTE
13 from sklearn.model_selection import GridSearchCV, StratifiedKFold
14
15 # Metrics
16 from sklearn.metrics import (
17     classification_report,
18     confusion_matrix,
19     roc_auc_score,
20     roc_curve,
21     average_precision_score,
22     f1_score,
23     recall_score,
24     precision_score
25 )
26
27 # Explainability
28 import shap
29
30 # Output directory for artifacts
31 OUTPUT_DIR = "Milestone_3"
32 os.makedirs(OUTPUT_DIR, exist_ok=True)
33
34 # Load preprocessed data from Milestone_1 folder
35 print("Loading preprocessed dataset...")
36 X_train = pd.read_csv("Milestone_1/X_train_scaled.csv")
```



```
1 import numpy as np
2 import pandas as pd
3 import joblib
4 import matplotlib.pyplot as plt
5 from sklearn.linear_model import LogisticRegression
6 from sklearn.metrics import classification_report, confusion_matrix, roc_auc_score
7
8 # =====
9 # 1. Load Preprocessed Datasets
10 # =====
11 X_train_scaled = pd.read_csv('Milestone_1/X_train_scaled.csv')
12 X_test_scaled = pd.read_csv('Milestone_1/X_test_scaled.csv')
13 y_train = pd.read_csv('Milestone_1/y_train.csv').values.ravel()
14 y_test = pd.read_csv('Milestone_1/y_test.csv').values.ravel()
15
16 # =====
17 # 2. Initialize Model & Fit on Training Data
18 # =====
19 model = LogisticRegression(class_weight='balanced', random_state=42, max_iter=1000)
20 model.fit(X_train_scaled, y_train)
21
22 # =====
23 # 3. Predict on Test Data
24 # =====
25 y_pred = model.predict(X_test_scaled) # Default 0.5 threshold
26 y_proba = model.predict_proba(X_test_scaled)[: , 1] # Failure probability scores
27
28 # =====
29 # 4. Print Confusion Matrix & Metrics
30 # =====
31 print("=== Confusion Matrix (Threshold 0.50) ===")
32 print(confusion_matrix(y_test, y_pred))
33
34 print("\n=== Classification Report ===")
35 print(classification_report(y_test, y_pred, zero_division=0))
36
```



```

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43
44 AI-Based-Predictive-Maintenance-and-Process-Intelligence-System/
45 |
46 |— README.md
47 |— ai4i2020.csv # Raw dataset (10,000 rows)
48 |— generate_ppt.py # PPT generation script
49 |
50 |— Milestone_1/ # Data Preprocessing & EDA
51 |   |— ai4i2020_preprocessing_eda.ipynb # Main EDA notebook
52 |   |— ai4i2020_clean.csv # Cleaned dataset
53 |   |— X_train_scaled.csv # Scaled training features
54 |   |— X_test_scaled.csv # Scaled test features
55 |   |— y_train.csv # Training labels
56 |   |— y_test.csv # Test labels
57 |
58 |— Milestone_2/ # Classical ML Models
59 |   |— logistic_regression.py # Logistic Regression
60 |   |— svm_code.py # Support Vector Machine
61 |   |— decision_tree.py # Decision Tree
62 |   |— logistic_regression_model.pkl # Saved LR model
63 |   |— svm_model.pkl # Saved SVM model
64 |   |— decision_tree_model.pkl # Saved DT model
65 |   |— feature_importance.png # LR odds-ratio chart
66 |
67 |— Milestone_3/ # Advanced Models
68 |   |— ensemble_models.py # Random Forest + Gradient Boosting
69 |   |— ann.py # ANN (MLPClassifier)
70 |   |— ann_predictive_maintenance.ipynb # ANN notebook
71 |   |— best_predictive_maintenance_model.pkl
72 |   |— ann_model.pkl
73 |   |— model_comparison.csv
74 |   |— roc_curves.png
75 |   |— confusion_matrices.png
76 |   |— shap_summary.png
77
78

```

7. Challenges Faced

During development of the AI-Based-Industrial-Maintenance-and-Process-Optimization, several technical and operational challenges were encountered:

Data preprocessing: Handling missing values, duplicate records, and converting categorical data into a suitable format for machine learning.

Imbalanced dataset: Machine failure cases were much fewer than normal cases, making failure prediction more challenging.

Feature selection: Identifying the most relevant machine parameters from the available features for accurate prediction.

Model performance: Different algorithms produced different results, so selecting the most suitable model required proper comparison.

False predictions: Reducing false positives and false negatives was challenging, especially because correctly detecting actual machine failures is important.

Model tuning: Finding suitable parameters for models such as Random Forest and XGBoost required testing different configurations.

Evaluation: Accuracy alone was not sufficient, so Precision, Recall, F1-score, and Confusion Matrix had to be considered together.

Process optimization: Converting machine-learning results into meaningful maintenance and process-improvement insights required careful analysis.

8. Learnings & Skills Acquired

The development of the AI-Based-Industrial-Maintenance-and-Process-Optimization resulted in significant technical, analytical, and professional learning outcomes:

1. Backend Development & Workflow Integration

Learned to design Django-based backend logic for handling file uploads, request routing, data flow, and output rendering.

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2. UI/UX Development & Dashboards

Enhanced skills in HTML, CSS, UI structuring, theme toggling, analytics visualization, and responsive layouts.

3. Debugging & Dependency Management

Encountered and solved version conflicts, library errors, and environment configuration challenges—strengthening troubleshooting skills.

4. System Testing & Iterative Improvement

Practiced evaluating output accuracy using sample datasets, identifying mismatch patterns, and fine-tuning algorithms.

5. Research & Skill Mapping Insight

Understood how industry job descriptions translate into competency requirements and how skill gaps inform learning recommendations.

6. Technical Communication & Documentation

Improved ability to document system architecture, progress tracking, reporting, and structured presentation delivery.

Overall, the project strengthened both **AI-technical competencies** and **software engineering mindset**, enhancing capability to build scalable, user-centric solutions.

9. Testimonials from team

Team Member 1

“This project deepened my understanding of **machine learning and predictive maintenance**, and exposed me to practical workflows for predicting industrial machine failures.”

Team Member 2

“This project strengthened my knowledge of **data preprocessing and exploratory data analysis**, and helped me understand how industrial data can be used for maintenance decisions.”

Team Member 3

“This project improved my understanding of **machine-learning model evaluation and comparison**, and gave me practical experience with classification algorithms such as Random Forest and XGBoost.”

Team Member 4

“This project enhanced my understanding of **feature importance and process optimization**, and showed me how AI can identify factors affecting industrial machine performance.”

Team Member 5

“This project strengthened my skills in **AI-based industrial problem solving**, and provided practical experience in developing a complete predictive maintenance workflow from data analysis to failure prediction.”

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10. Conclusion

The **AI-Based Industrial Maintenance and Process Optimization** project successfully demonstrates the application of Artificial Intelligence and Machine Learning in predicting industrial machine failures and supporting preventive maintenance. The project analyzed machine parameters such as temperature, rotational speed, torque, and tool wear to identify patterns associated with machine failures.

Multiple machine-learning models were implemented and evaluated using Accuracy, Precision, Recall, F1-score, and Confusion Matrix to identify the most suitable model for failure prediction. Feature-importance

analysis also helped identify the key parameters influencing machine failure.

Overall, the system provides a data-driven approach to **early failure detection, preventive maintenance, reduction of unexpected downtime, and process optimization**. The project demonstrates how AI and machine learning can improve industrial maintenance planning and support more efficient and reliable production processes.

11. Acknowledgements

We express our sincere gratitude to **Infosys Springboard** for providing us with an excellent learning environment, industry-level curriculum support, and access to resources that helped shape this project effectively.

We extend heartfelt thanks to **Ms. Sangeetha**, our mentor, for her continuous guidance, constructive feedback, and encouragement throughout the development lifecycle. Her insights played a pivotal role in refining our approach, validating our methods, and enhancing our learning experience.

We also appreciate the support from our faculty members, peers, and family for their motivation, collaboration, and valuable suggestions, which helped us stay committed and deliver the AI-Based-Industrial-Maintenance-and-Process-Optimization project successfully